# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 7 — SCIENCE UNIT TEST SET A

Chapters 1–4: **Exploring Substances: Acidic, Basic, and Neutral, Electricity: Circuits and Their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

## SECTION — A (Objective Type Questions) 12 × 1 = 12

*Choose the correct option and write it in your answer book.*

**1.** The reaction between an acid and a base is called: 1

**(a)** neutralisation **(b)** rusting **(c)** combustion **(d)** evaporation

**2.** Which of these is basic? 1

**(a)** Soap solution **(b)** Amla juice **(c)** Vinegar **(d)** Tamarind water

**3.** The acid present in tamarind is: 1

**(a)** lactic acid **(b)** citric acid **(c)** tartaric acid **(d)** acetic acid

**4.** In the symbol of a cell, the longer line represents the: 1

**(a)** bulb **(b)** negative terminal

**(c)** positive terminal **(d)** switch

**5.** An electrical component that protects circuits by melting is a: 1

**(a)** bulb **(b)** cell **(c)** switch **(d)** fuse

**6.** When you press an electric bell, the circuit gets: 1

**(a)** open **(b)** closed **(c)** short only **(d)** cut

**7.** The oxides of non-metals are generally: 1

**(a)** neutral **(b)** salty **(c)** acidic **(d)** basic

**8.** Which non-metal is a liquid at room temperature? 1

**(a)** Bromine **(b)** Mercury **(c)** Sulfur **(d)** Carbon

**9.** The black coating on silver objects is called: 1

**(a)** rust **(b)** verdigris **(c)** tarnish **(d)** lustre

**10.** Milk turning into curd is a: 1

**(a)** evaporation **(b)** physical change

**(c)** chemical change **(d)** reversible change

**11.** A candle melting is ___, while the burning wick is ___. 1

**(a)** chemical, physical **(b)** physical, chemical

**(c)** physical, physical **(d)** chemical, chemical

**12.** The lowest temperature at which a substance catches fire is its: 1

**(a)** boiling point **(b)** room temperature

**(c)** ignition temperature **(d)** melting point

## SECTION — B (Very Short Answer Type) — 5 × 2 = 10

*Answer in about 15–25 words each.*

**13.** How would you test whether a solution is acidic using litmus paper? 2

**14.** Why does a bulb not glow when the switch is open? 2

**15.** What is a circuit diagram? 2

**16.** Name one liquid metal and one liquid non-metal. 2

**17.** What is combustion? Give two combustible substances. 2

## SECTION — C (Short Answer Type) — 6 × 3 = 18

*Answer in about 40–60 words each.*

**18.** A student tests four solutions: P turns blue litmus red; Q turns red litmus blue; R no change; S turns blue litmus red. Classify. 3

**19.** A bulb does not glow even after all connections are made. List three possible reasons. 3

**20.** Describe an activity to show metals are good conductors of heat. 3

**21.** Classify the changes: melting of butter, burning of candle, cooking egg, dissolving salt. 3

**22.** Why is rusting harmful? How can it be prevented? 3

OR

Describe what happens when a candle burns, pointing out physical and chemical changes. 3

**23.** What is the importance of acids and bases in the digestive system? 3

OR

Explain why a bee sting is treated differently from a wasp sting. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 80–120 words each.*

**24.** (i) What are the effects of electric current? 5
(ii) Which one is seen in a bulb?
(iii) Which in a heater?

OR

**Case study:** A student tests lemon juice, sugar solution, salt solution and distilled water in a tester. The bulb glows for lemon and salt only. 5
(i) Conclusion?
(ii) Why not for sugar?
(iii) Why not distilled water?
(iv) Why add salt to water?

**25.** (i) Explain rusting and what is needed. 5
(ii) Describe an experiment.
(iii) State methods of prevention.

OR

**Case study:** Priya mixes vinegar with baking soda and passes the gas through lime water. 5
(i) What do you see?
(ii) Name the gas.
(iii) What happens to lime water?
(iv) Is it a chemical change?

## END OF QUESTION PAPER — Total: 50 Marks